

Haoran (Matt) Wan

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RESEARCH INTERESTS

My research program examines decision-making across the adult lifespan, with a specific focus on the psychological mechanisms underlying how individuals discount the value of future and uncertain outcomes. Using large-scale experiments and advanced computational modeling, my work investigates how factors such as age and socioeconomic status shape everyday choices. Specifically, I test two overarching hypotheses: a buffering hypothesis, which posits that age-related psychological factors buffer older adults against the negative effects of financial scarcity, and a complexity hypothesis, which predicts that age differences are amplified in more complex, multi-attribute decisions. The ultimate goal of this work is to build more nuanced predictive models of financial and health-related decision-making.

EDUCATION

WASHINGTON UNIVERSITY IN ST. LOUIS, St. Louis, MO

Ph.D. in Psychological & Brain Sciences Expected 2026

Graduate Certificate in Quantitative Data Analysis December 2023

M.A. Psychological & Brain Sciences December 2023

REED COLLEGE, Portland, OR

B.A. Economics & Psychology May 2021

AWARDS & GRANTS

University Fellowship, Washington University in St. Louis 2021 – Present

Sallie P. Asche Travel Awards, Center for Vital Longevity 2023, 2025

Tony Nevin Student Award, Society for the Quantitative Analyses of Behavior 2022

Gerald M. Meier Award for Distinction in Economics, Reed College 2021

Career Advancement Fund, Reed College 2020

Student Summer Research Fellowship, Reed College 2020

Opportunity Grants, Reed College 2017, 2018

Commendation for Excellence, Reed College 2019, 2020, 2021

POSITIONS HELD

Research

Graduate Student Researcher, Washington University in St. Louis, St. Louis, MO 2021 – Present

PI: Drs. Leonard Green & Joel Myerson, Behavioral Economics Lab

Research Assistant, Reed College, Portland, OR 2020

PI: Dr. Denise Hare, Economics Department

Research Assistant, Reed College, Portland, OR 2017 – 2021

PI: Dr. Timothy Hackenberg, Learning & Adaptive Behavior Lab

Teaching

<i>Graduate Teaching Assistant</i> , Washington University in St. Louis	2025
PSYCH 5068 Hierarchical Linear Models, Dr. Mike J Strube	
<i>Lecturer</i> , Washington University in St. Louis	2024
PSYCH 102 First-Year Opportunity: Contemporary Issues in Psychology	
<i>Assistant to Instructor</i> , Washington University in St. Louis	2022 – 2024
PSYCH 3890 Advanced Psychological Statistics, Drs. Shelly Cooper & Joshua Jackson	
PSYCH 5068 Hierarchical Linear Models, Dr. Mike J Strube	
PSYCH 361 Psychology of Learning, Dr. Leonard Green	
<i>Teaching Assistant</i> , Reed College	2020 – 2021
ECON 201 Introduction to Economic Analysis, Dr. Denise Hare	
PSYCH 203 Learning & Comparative Psychology, Dr. Timothy Hackenberg	
PSYCH 373 Learning, Dr. Timothy Hackenberg	

Teaching Interests

I am prepared to teach a range of undergraduate and graduate courses in quantitative methods and behavioral science. My teaching approach emphasizes rigorous methodology, critical thinking, and the application of theoretical concepts to real-world problems.

Quantitative Methods

Advanced Statistics/Data Analysis (Graduate)
 Hierarchical Linear Models (Graduate)
 Applied Bayesian Statistics (Graduate)
 Longitudinal Data Analysis (Graduate)
 Research Methods & Experimental Design

Behavioral & Decision Science

Introduction to Psychology/Behavioral Science
 Judgment and Decision-Making
 Psychology of Aging
 Psychology of Learning
 Health Decision-Making

Internship

<i>Software Engineer</i> , Swipesum, St. Louis, MO	Fall 2025
<i>Data Scientist</i> , Swipesum, St. Louis, MO	Summer 2025
<i>Quantitative Medicine Analyst</i> , Critical Path Institute, Tucson, AZ	Summer 2024
<i>Legal Analyst; Fixed-Income Analyst</i> , Shenzhen Stock Exchange, Shenzhen, China	Summer 2019

PUBLICATIONS

- Wan, H.**, Myerson, J., Green, L., Strube, M. J., & Hale, S. (2025). Age, income, and the discounting of delayed monetary losses. *The Journals of Gerontology: Series B, Psychological Sciences and Social Sciences*, Article gbaf162. <https://doi.org/10.1093/geronb/gbaf162>
- Oliveira, L., Green, L., & Myerson, J., & **Wan, H.** (2025). Discounting of probabilistic food reinforcement by pigeons. *Journal of the Experimental Analysis of Behavior*, 124(1). <https://doi.org/10.1002/jeab.70042>
- Wan, H.**, Green, L., & Myerson, J. (2024). Delayed monetary losses: Do different procedures and discounting measures assess the same construct?. *Behavioural Processes*, 105101. <https://doi.org/10.1016/j.beproc.2024.105101>

- Wan, H.,** Myerson, J., Green, L., Strube, M. J., & Hale, S. (2024). Age-related differences in delay discounting: Income matters. *Psychology and Aging*, 39(6), 632.
<https://doi.org/10.1037/pag0000818>
- Wan, H.** (2023). *Discounting of delayed and probabilistic outcomes across the adult lifespan*. (Master's Thesis, Washington University in St. Louis).
- Schulinkamp, R., **Wan, H.,** & Hackenberg, T. D. (2023). Social familiarity and reinforcement value: A behavioral-economic analysis of demand for social interaction with cagemate and non-cagemate female rats. *Frontiers in Psychology*, 14, 1158365.
<https://doi.org/10.3389/fpsyg.2023.1158365>
- Wan, H.,** Myerson, J., & Green, L. (2023). Individual differences in degree of discounting: Do different procedures and measures assess the same construct? *Behavioural Processes*, 104864.
<https://doi.org/10.1016/j.beproc.2023.104864>
- Kirkman, C., **Wan, H.,** & Hackenberg, T. D. (2022). A behavioral-economic analysis of demand and preference for social and food reinforcement in rats. *Learning and Motivation*, 77, 101780.
<https://doi.org/10.1016/j.lmot.2021.101780>
- Wan, H.,** Kirkman, C. F., Jensen, G., Hackenberg, T. D. (2021). Failure to find altruistic food sharing in rats. *Frontiers in Psychology*, 12:696025. <https://doi.org/10.3389/fpsyg.2021.696025>

MANUSCRIPTS UNDER REVIEW

- Wan, H.,** Myerson, J., & Green, L. (*under review*). Brief assessments of delay discounting: Two-amount monetary choice and delayed losses questionnaires.
- Myerson, J., Green, L., Vanderveldt, A., & **Wan, H.** (*under review*). Delay-probability asymmetry in discounting: An anomaly in multi-attribute choice.
- Wan, H.,** Tan, L., & Hackenberg, T. (*under review*). Behavioral economic analysis of pigeons' token accumulation and reinforcer demand in a laboratory-based token economy.

MANUSCRIPTS IN PREPARATION

- Wan, H.** (*in preparation*). Discounting of rewards that are both delayed and probabilistic: The effects of age and financial stress.
- Wan, H.,** Myerson, J., Green, L., Strube, M. J., & Hale, S. (*in preparation*). Age, income, and the discounting of delayed and probabilistic rewards.
- Wan, H.,** Myerson, J., Green, L. (*in preparation*). A parsimonious simplification of killean's general theory of discounting.
- Wan, H.,** Myerson, J., Green, L. (*in preparation*). A Multiplicative Relation: On the Independence of Delay and Probability Discounting.

CONFERENCE PRESENTATIONS

- Wan, H.,** Green, L., & Myerson, J. (2025, May). Brief assessments of delay discounting: Two-amount monetary choice and delayed losses questionnaires [Poster presentation]. 51st Annual Convention of the Association for Behavior Analysis International, Washington, D.C.

- Wan, H.,** Green, L., Myerson, J., Strube, M., Hale, S., & Chen, D. (2025, February). Age, income, and the discounting of delayed monetary losses [Poster presentation]. 8th Dallas Aging and Cognition Conference, Dallas, TX.
- Wan, H.,** Green, L., & Myerson, J. (2024, May). Discounting delayed losses: Do different procedures and measures assess the same construct? [Poster presentation]. 50th Annual Convention of the Association for Behavior Analysis International, Philadelphia, PA.
- Wan, H.,** Myerson, J., Green, L., & Strube, M. (2024, May). Discounting of rewards that are both delayed and probabilistic: A comparison of models [Poster presentation]. 46th Annual Meeting of the Society for the Quantitative Analyses of Behavior, Philadelphia, PA.
- Wan, H.,** Green, L., & Myerson, J. (2023, May). Changes in discounting of gains and losses across adulthood [Poster presentation]. 49th Annual Convention of the Association for Behavior Analysis International, Denver, CO.
- Wan, H.,** Schulingkamp, R., & Hackenberg, T. (2023, May). Social familiarity and reinforcement value: A behavioral-economic analysis of demand for social interaction in rats with familiar and less familiar social partners [Poster presentation]. 45th Annual Meeting of the Society for the Quantitative Analyses of Behavior, Denver, CO.
- Wan, H.,** Green, L., Myerson, J., Strube, M., & Hale, S. (2023, February). The effects of age and income on discounting of delayed rewards [Poster presentation]. 7th Dallas Aging and Cognition Conference, Dallas, TX.
- Rosenthal, D., Green, L., & **Wan, H.** (2023, April). Do you (scarcity) mind(set)? The effect of subjective scarcity on decision-making [Poster presentation]. Spring 2023 Undergraduate Research Symposium, St. Louis, MO.
- Wan, H.** (2022, April). The effects of age and income on the discounting of delayed rewards [Talk presentation]. Brain, Behavior, and Cognition Brown Bag Series, St. Louis, MO.
- Wan, H.,** Myerson, J., & Green, L. (2022, May). Comparison of the adjusting-amount procedure and the monetary choice questionnaire for measuring delay discounting [Talk presentation]. 44th Annual Meeting of the Society for the Quantitative Analyses of Behavior, Boston, MA.
- Cao, A., Tan, L., **Wan, H.,** & Hackenberg, T. D. (2022, May). Own- and cross-price demand elasticity with specific and generalized conditioned reinforcers in a token economy with pigeons [Poster presentation]. 44th Annual Meeting of the Society for the Quantitative Analyses of Behavior, Boston, MA.
- Wan, H.** (2021, December). Comparison of the adjusting-amount procedure and the monetary choice questionnaire for measuring delay discounting [Talk presentation]. Brain, Behavior, and Cognition Brown Bag Series, St. Louis, MO.
- Wan, H.,** Tan, L., & Hackenberg, T. D. (2021, May). Economic analysis of pigeons' token production, exchange, and accumulation in a laboratory-based token economy [Poster presentation]. 43rd Annual Meeting of the Society for the Quantitative Analyses of Behavior (Virtual).
- Wan, H.,** Kirkman, C. F., Franceschini, C., & Hackenberg, T. D. (2019, May). Failure to find altruistic behavior in rats [Poster presentation]. 45th Annual Convention of the Association for Behavior Analysis International, Chicago, IL.

- Hackenberg, T. D., **Wan, H.**, & Kirkman, C. F. (2019, May). Do rats share food? Tests with the social-release paradigm [Paper presentation]. 45th Annual Convention of the Association for Behavior Analysis International, Chicago, IL.
- Kirkman, C. F., **Wan, H.**, Franceschini, C., & Hackenberg, T. D. (2019, May). Assessing cross-price interactions between food and social reinforcement [Poster presentation]. 45th Annual Convention of the Association for Behavior Analysis International, Chicago, IL.
- Hackenberg, T. D., Kirkman, C. F., **Wan, H.**, & Franceschini, C. (2019, September). Behavioral economics of food and social reinforcement [Paper presentation]. 10th International Meeting of the Association for Behavior Analysis International, Stockholm, Sweden.
- Wan, H.**, Kirkman, C. F., Franceschini, C., & Hackenberg, T. D. (2018, April). Failure to find altruistic behavior in rats [Poster presentation]. Annual Research Poster Session, Reed College, Portland, OR.

PROFESSIONAL AFFILIATIONS

Association for Psychological Science
 Association for Behavior Analysis International
 Society for the Quantitative Analyses of Behavior
 APA Division 25 Behavior Analysis

REVIEWING

Journal of the Experimental Analysis of Behavior
Behavioural Processes

PROFESSIONAL ACTIVITIES

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| <i>Member</i> , Pivot 314. The Graduate Center, Washington University in St. Louis. | 2025 |
| <i>Attendee</i> , Issues in Aging Seminar Series. The Harvey A. Friedman Center for Aging, Washington University in St. Louis. | 2024, 2025 |
| <i>Attendee</i> , Computational Modeling Workshop. Scientific Research Network on Decision Neuroscience and Aging. | 2024 |
| <i>Judge</i> , 29 th Graduate Student Research Symposium. Graduate Student Senate, Washington University in St. Louis. | 2024 |
| <i>Attendee</i> , Joint FDA/FNIH Workshop for Digital Measures. Foundation for the National Institutes of Health | 2024 |
| <i>Volunteer</i> , Society for the Quantitative Analyses of Behavior. | 2023 |
| <i>Chair</i> , Graduate Student Brown Bag Series. Department of Psychological & Brain Sciences, Washington University in St. Louis. | 2022 – 2023 |
| <i>Member</i> , Washington University Chinese Students and Scholars Association. Washington University in St. Louis. | 2022 – 2023 |
| <i>Attendee & Presenter</i> , Brain, Behavior, and Cognition Brown Bag Series. Department | 2021 – 2025 |

of Psychological and Brain Sciences, Washington University in St. Louis.
Attendee, Department Colloquia Series. Department of Psychological and Brain Sciences, Washington University in St. Louis. 2021 – 2025
Presenter, Dallas Aging and Cognition Conference. Dallas, TX. 2023, 2025
Presenter, Society for the Quantitative Analysis of Behavior Annual Conference 2019 – 2025
Presenter, Association for Behavior Analysis International Annual Convention 2019, 2021-2025

MENTORING

Mars, W. (2025). *Discounting of Probabilistic Losses Across the Adult Lifespan*. (Research Project, Washington University in St. Louis).
Chen, D. (2025). *Age, Income, and the Discounting of Delayed Monetary Losses*. (Research Project, Washington University in St. Louis).
Rosenthal, D. (2023). *Do You (Scarcity) Mind(set)?: A Subjective Examination of Scarcity Theory*. (Senior Honors Thesis, Washington University in St. Louis).
Sheldon, M. (2022). *The Relation Between Probability and Delay Discounting: What is the Immediate Probabilistic Value of a Delayed Reward?* (Honors Thesis, Washington University in St. Louis).

REFERENCE

Leonard Green, Ph.D.

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Professor of Economics
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Joel Myerson, Ph.D.

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Michael Strube, Ph.D.

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